

CBSE 2027 Physics — Predicted Paper (Set 10 of 10)

SECTION A: Objective Type Questions (16 Marks)

This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.

Q1. An isolated point charge particle produces an electric field $\vec{E}$ at a point **3 m** away from it. The distance of the point at which the field is $\vec{E}/4$ will be: (a) **2 m** (b) **3 m** (c) **4 m** (d) **6 m**
(Chapter: Electric Charges and Fields) | **[Confidence: Very High]**

Q2. Beams of electrons and protons move parallel to each other in the exact same direction. They will: (a) attract each other (b) repel each other (c) neither attract nor repel (d) experience a force depending upon the speed of the beams (Chapter: Moving Charges and Magnetism) | **[Confidence: Very High]**

Q3. Three charges $2q$, $-q$, and $-q$ lie at the three vertices of an equilateral triangle. The value of the net electric field $\vec{E}$ and net electric potential V at the centroid of the triangle will be: (a) $\vec{E} \neq 0$ and $V \neq 0$ (b) $\vec{E} = 0$ and $V = 0$ (c) $\vec{E} \neq 0$ and $V = 0$ (d) $\vec{E} = 0$ and $V \neq 0$
(Chapter: Electrostatic Potential and Capacitance) | **[Confidence: High]**

Q4. A potential difference of **200 V** is maintained across a conductor of resistance **100 Ω** . The number of electrons passing through it in **1 s** is: (a) 1.25×10^{19} (b) 2.5×10^{18} (c) 1.25×10^{10} (d) 2.5×10^{16} (Chapter: Current Electricity) | **[Confidence: High]**

Q5. An electron with angular momentum L moving around the nucleus has a magnetic dipole moment given by: (a) $eL/2m$ (b) $eL/3m$ (c) $eL/4m$ (d) eL/m (Chapter: Magnetism and Matter / Atoms) | **[Confidence: Very High]**

Q6. If the magnetizing field applied to a ferromagnetic material is steadily increased, its magnetic permeability: (a) decreases (b) increases (c) remains unchanged (d) first decreases and then increases (Chapter: Magnetism and Matter) | **[Confidence: Very High]**

Q7. A coil of area **100 cm²** is kept at an angle of **30°** with a magnetic field of **10⁻¹ T**. The magnetic field is reduced to zero in **10⁻⁴ s**. The induced emf in the coil is: (a) $5\sqrt{3}$ V (b) $50\sqrt{3}$ V (c) **5.0 V** (d) **50.0 V** (Chapter: Electromagnetic Induction) | **[Confidence: High]**

Q8. The rms current in a circuit connected to a **50 Hz** AC source is **15 A**. The value of the current in the circuit $\frac{1}{600}$ s after the instant the current is zero, is: (a) $15/\sqrt{2}$ A (b) $15\sqrt{2}$ A (c) $\sqrt{2}/15$ A (d) $15\sqrt{3/2}$ A (Chapter: Alternating Current) | **[Confidence: Medium]**

Q9. A ray of monochromatic light of wavelength **600 nm** propagates from air into a medium. If its wavelength in the medium becomes **400 nm**, the refractive index of the medium is: (a) **1.4** (b) **1.5** (c) **1.6** (d) **1.8** (Chapter: Ray Optics and Optical Instruments) | **[Confidence: Very High]**

Q10. In a Young's double-slit experiment, the path difference at a certain point on the screen between two interfering waves is $\frac{1}{8}^{th}$ of the wavelength ($\lambda/8$). The ratio of intensity at this

point to the maximum intensity at the centre of a bright fringe is close to: (a) **0.80** (b) **0.74** (c) **0.94** (d) **0.85** (*Chapter: Wave Optics*) | **[Confidence: High]**

Q11. A photon of wavelength **663 nm** is incident on a metal surface. The work function of the metal is **1.50 eV**. The maximum kinetic energy of the emitted photoelectrons is: (a) **3.0×10^{-20} J** (b) **6.0×10^{-20} J** (c) **4.5×10^{-20} J** (d) **9.0×10^{-20} J** (*Chapter: Dual Nature of Radiation and Matter*) | **[Confidence: Very High]**

Q12. Specify the transition of an electron in the Bohr model of a hydrogen atom which gives rise to the spectral line of the highest wavelength in the Paschen series. (a) **$n = 3$ to $n = 1$** (b) **$n = 3$ to $n = 2$** (c) **$n = 4$ to $n = 1$** (d) **$n = 4$ to $n = 3$** (*Chapter: Atoms*) | **[Confidence: Very High]**

Q13. Assertion (A): An electron has a higher potential energy when it is at a location associated with a more negative value of electric potential. **Reason (R):** Electrons naturally move from a region of higher potential to a region of lower potential. (*Chapter: Electrostatic Potential and Capacitance*) | **[Confidence: High]**

Q14. Assertion (A): The higher the range of an ammeter, the lower is its overall electrical resistance. **Reason (R):** To immensely increase the range of an ammeter, an additional shunt resistance is added in series to it. (*Chapter: Moving Charges and Magnetism*) | **[Confidence: Very High]**

Q15. Assertion (A): A convex mirror can never form a real image, regardless of the object's position. **Reason (R):** A convex mirror converges the parallel rays that are incident on it. (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: High]**

Q16. Assertion (A): The photoelectrons produced by a monochromatic light beam incident on a metal surface have a spread in their kinetic energies. **Reason (R):** The energy of electrons emitted from inside the bulk of the metal surface is partially lost in collisions with the other atoms in the metal before escaping. (*Chapter: Dual Nature of Radiation & Matter*) | **[Confidence: Very High]**

SECTION B: Very Short Answer Type Questions (10 Marks)

This section consists of 5 questions of 2 marks each.

Q17. Two point charges of **$+1 \mu\text{C}$** and **$+4 \mu\text{C}$** are kept **30 cm** apart. How far from the **$+1 \mu\text{C}$** charge, on the straight line joining the two charges, will the net electric field be strictly zero? (*Chapter: Electric Charges and Fields*) | **[Confidence: Very High]**

Q18. Two bulbs are rated as **(P_1, V)** and **(P_2, V)** . If they are connected (i) in series and (ii) in parallel across a supply voltage **V** , mathematically find the total power dissipated in the two combinations in terms of **P_1** and **P_2** . (*Chapter: Current Electricity*) | **[Confidence: High]**

Q19. A closely wound solenoid of **2000** turns and an area of cross-section **$1.6 \times 10^{-4} \text{ m}^2$** , carrying a current of **4.0 A**, is suspended through its center allowing it to turn in a horizontal

plane. What is the magnetic dipole moment associated with this solenoid? (*Chapter: Magnetism and Matter*) | **[Confidence: Very High]**

Q20. During the charging of a capacitor, the steady conduction current in the connecting wires is **0.25 A**. What is the magnitude of the displacement current across its plates? State the underlying principle justifying your answer. (*Chapter: Electromagnetic Waves*) | **[Confidence: Very High]**

Q21. If the nuclear radius of Aluminum ${}_{13}^{27}\text{Al}$ is experimentally found to be **3.6 fermi**, find the nuclear radius of Copper ${}_{29}^{64}\text{Cu}$. (*Chapter: Nuclei*) | **[Confidence: Very High]**

SECTION C: Short Answer Type Questions (21 Marks)

This section consists of 7 questions of 3 marks each.

Q22. A parallel plate capacitor (A) of capacitance C is fully charged by a battery to a voltage V . The battery is then permanently disconnected, and a completely uncharged capacitor (B) of capacitance $2C$ is connected in parallel across A. Find the ratio of: (i) The final charges residing on capacitor A and B. (ii) The total electrostatic energy stored in A and B finally to the energy stored in A initially. (*Chapter: Electrostatic Potential and Capacitance*) | **[Confidence: Very High]**

Q23. A potential difference V is applied across the ends of a uniform copper wire of length l and diameter D . Using the relevant mathematical expression, explain the exact effect on the drift velocity of the electrons if: (i) The applied voltage V is halved. (ii) The length l is doubled. (iii) The diameter D is halved. (*Chapter: Current Electricity*) | **[Confidence: Very High]**

Q24. The current flowing in two coils of self-inductances $L_1 = 16 \text{ mH}$ and $L_2 = 12 \text{ mH}$ is increasing at the exact same rate. If the instantaneous power supplied to the two coils is equal, find the ratio of: (i) the induced voltage in the coils, (ii) the currents flowing in the coils, (iii) the magnetic energies stored in the two coils at that given instant. (*Chapter: Electromagnetic Induction*) | **[Confidence: High]**

Q25. An equilateral glass prism has a refractive index of **1.6** in air. Calculate the angle of minimum deviation of this prism when it is fully immersed in a liquid medium of refractive index $\frac{4\sqrt{2}}{5}$. (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Very High]**

Q26. (i) In a Young's Double Slit Experiment using light of wavelength **600 nm**, the angular width of the fringe formed on a distant screen is observed to be **0.1°**. Find the physical spacing between the two slits. (ii) If a light wave of wavelength **5000 Å** propagating in air gets partly reflected and partly refracted from the surface of water, how will the wavelengths and frequencies of the reflected and refracted light be affected? (*Chapter: Wave Optics*) | **[Confidence: High]**

Q27. A monochromatic light of frequency $6.0 \times 10^{14} \text{ Hz}$ is produced by a laser. The total power emitted by the source is $2.0 \times 10^{-3} \text{ W}$. Calculate: (i) the energy of a single photon in

the light beam, and (ii) the total number of photons emitted on an average per second by the source. (*Chapter: Dual Nature of Radiation and Matter*) | **[Confidence: Very High]**

Q28. Name the specific extrinsic semiconductors formed when a pure Germanium crystal is doped independently with (i) a trivalent impurity, and (ii) a pentavalent impurity. Draw the characteristic energy band diagrams of the two extrinsic semiconductors so formed, clearly showing the donor/acceptor energy levels. (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]**

SECTION D: Case-Study Based Questions (8 Marks)

This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts.

Q29. Case Study 1: V-I Characteristics of a Semiconductor Diode *Passage:* The circuit arrangement for studying the V-I characteristics of a p-n junction diode involves measuring current as a function of the applied voltage. In forward bias measurement, a milliammeter is used since the expected current is large due to majority carrier diffusion, while a microammeter is used in reverse bias to measure the small drift current. In forward bias, the current first increases very slowly, almost negligibly, till the voltage across the diode crosses a certain characteristic value. After this characteristic voltage, the diode current increases significantly (exponentially) for even a very small increase in the bias voltage. (a) What is the specific term used for the characteristic forward voltage at which the current increases rapidly? Write its approximate value for a Silicon diode. (1 mark) (b) Explain the origin of the potential barrier in an unbiased p-n junction. What specific charges create it? (1 mark) (c) How does the dynamic electrical resistance of the semiconductor diode change as the forward bias voltage is increased beyond the cut-in voltage? **OR** Why is the net junction current perfectly zero in an unbiased p-n junction at equilibrium? Explain in terms of drift and diffusion currents. (2 marks) (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]**

Q30. Case Study 2: Optical Submersion and Total Internal Reflection *Passage:* A point source of light is placed at the bottom of a tank filled with water of refractive index μ to a depth d . As light rays emanate from the source and hit the water-air interface, they refract away from the normal. As the angle of incidence increases, the angle of refraction increases until the refracted ray grazes the surface. Beyond this critical angle, total internal reflection traps the light within the denser medium. Thus, the light can only emerge from the water surface through a specific circular area directly above the source. (a) Define the critical angle for a pair of given media. (1 mark) (b) Out of Crown glass, Flint glass, Water, and Diamond, which transparent medium has the minimum critical angle with respect to air? (1 mark) (c) Derive the mathematical expression for the area of the circular surface of water through which the light from the submerged point source can emerge, in terms of d and μ . **OR** The critical angle for a pair of two media A and B of refractive indices 2.0 and 1.0 respectively is to be determined. Calculate this critical angle and the speed of light in medium A. (2 marks) (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Very High]**

SECTION E: Long Answer Type Questions (15 Marks)

This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.

Q31. (a) State Gauss's Theorem in electrostatics. **(b)** A right circular cylinder of length **20 cm** and radius **5 cm** has its center at the origin and its axis along the x-axis so that one flat face is at $x = +10 \text{ cm}$ and the other is at $x = -10 \text{ cm}$. An electric field is uniform and acts along the +x direction in the region of positive x, and is uniform with the same magnitude but acts in the -x direction in the region of negative x. The value of the field is $\vec{E} = 200\hat{i} \text{ N/C}$ for $x > 0$ and $\vec{E} = -200\hat{i} \text{ N/C}$ for $x < 0$. Calculate: (i) The net outward electric flux through the entire cylinder. (ii) The total net charge present identically inside the cylinder. **OR (a)** Using Gauss's Law, derive the expression for the electric field due to a uniformly charged thin spherical shell of radius R at a point P distant r from its center ($r > R$). **(b)** Draw a graph showing the variation of electric field E with distance r from the center of the shell for both $r < R$ and $r > R$. **(c)** Two concentric, hollow, thin spherical shells S_1 and S_2 of radii R and $2R$ enclose charges Q and $2Q$ respectively. What is the ratio of the electric flux passing through S_1 and S_2 ? (Chapter: Electric Charges and Fields) | **[Confidence: Very High]**

Q32. (a) Draw a neat schematic diagram of an AC Generator. **(b)** State its working principle and rigorously obtain the expression for the instantaneous value of the induced EMF (ϵ) in terms of the magnetic field B , number of turns N of the coil, area of cross-section A , and the angular frequency ω . **(c)** A circular coil of radius **10 cm** and 20 turns is rotated about its vertical diameter with an angular speed of **50 rad s⁻¹** in a uniform horizontal magnetic field of **3.0 × 10⁻² T**. Calculate the maximum EMF induced in the coil. **OR (a)** Explain with the help of a labelled diagram the fundamental principle and working of a step-down transformer. **(b)** State the underlying assumptions of an "ideal" transformer and deduce the ratio relation between the primary and secondary voltages and currents. **(c)** The primary coil of an ideal step-up transformer has 100 turns and the transformation ratio is also 100. The input voltage and power are respectively **220 V** and **1100 W**. Calculate: (i) the number of turns in the secondary, and (ii) the current flowing in the secondary circuit. (Chapter: Electromagnetic Induction / Alternating Current) | **[Confidence: Very High]**

Q33. (a) A point object O is kept in a rarer medium of refractive index n_1 in front of a convex spherical refracting surface of radius of curvature R , which separates a denser medium of refractive index n_2 from the rarer medium. Draw the ray diagram showing the image formation and deduce the relationship between the object distance (u) and the image distance (v) in terms of n_1 , n_2 , and R . **(b)** If a similar point object is placed in air symmetrically at a distance of **60 cm** in front of a concave spherical glass surface (refractive index **1.5**) having a radius of curvature of **20 cm**, find the exact position of the final image formed. **OR (a)** Why cannot the phenomenon of interference be observed by illuminating two separate, independent pinholes with two identical sodium lamps? Explain the concept of coherent sources. **(b)** Two monochromatic waves having displacements $y_1 = a \cos(\omega t)$ and $y_2 = a \cos(\omega t + \phi)$ from two coherent sources interfere to produce an interference pattern. Derive the expression for the resultant intensity and mathematically obtain the phase

conditions for constructive and destructive interference. (*Chapter: Ray Optics / Wave Optics*) |
[Confidence: Very High]

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.